

Weekly Research Report — Week 6

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Summary of Work Done

The evaluation was scaled from the 20-question pilot to the complete 1,273-question MedQA-USMLE test set, with paired significance testing. The findings reversed at scale and the paper was rewritten accordingly. A knowledge-graph scaling experiment and a domain subset analysis were added.

Full-scale evaluation

All five architectures evaluated on all 1,273 test questions, identical question set and ordering across systems so that every comparison is paired.

Architecture	Accuracy	vs baseline
Baseline LLM, no retrieval	77.1%	—
Flat-vector RAG	76.7%	$p = 0.65$, n.s.
Static GraphRAG	73.4%	$p < 0.001$
Agentic GraphRAG, self-correction	73.1%	$p < 0.001$
Agentic GraphRAG, dynamic PubMed	74.2%	$p = 0.001$

The pilot findings did not survive scaling. At 20 questions flat-vector RAG appeared to degrade performance by 5 points and agentic methods appeared to gain 5. At full scale flat-vector RAG is statistically indistinguishable from the baseline, and all three graph-based systems are significantly *worse* than it.

The no-retrieval baseline is the strongest system. This is the opposite of what the hypotheses predicted.

Knowledge-graph scaling experiment

To test whether the graph-based deficit was caused by an under-sized graph, the knowledge graph was rebuilt at twenty times the original scale — from 50 source documents to 1,000, yielding 3,489 entities. Both graph sizes were evaluated under the static and agentic architectures.

The deficit did not recover. Graph size is not the bottleneck. The more likely explanation is that the injected context is redundant when the base model already encodes the benchmark’s knowledge, and redundant context displaces correct parametric reasoning.

Domain subset analysis

Following the suggestion to examine domain-specific behaviour, 598 pharmacology-related questions were identified by keyword filtering and all five systems re-evaluated on that subset. Pharmacology is the natural candidate: drug indications, contraindications and interactions are highly relational and continuously updated, so it should favour both graph structure and live literature retrieval.

Every system performs marginally better on the subset (+1.4 to +2.3 points), indicating the questions are somewhat easier rather than that any architecture benefits disproportionately. The baseline remains strongest at 78.6%. Static and agentic GraphRAG remain significantly below it.

One effect is genuine: the dynamic PubMed system’s significant deficit against the baseline **disappears within pharmacology** (76.4% vs 78.6%, $p = 0.12$). Live literature retrieval is most competitive precisely where current structured knowledge is densely represented in the retrieved evidence.

Writing and housekeeping

- Abstract, Introduction, Methodology, Results and Conclusion rewritten to report the negative finding accurately rather than the hypothesised one
- Reference audit: preprint citations replaced with peer-reviewed versions where available (Nature Medicine, EMNLP 2024, ICML 2023); in-text citation numbering corrected
- Results tables converted to figures; methodology diagram tidied; scoping-review mind map revised with reference numbers per feedback
- Explicit hypothesis-to-metric mapping added

- Limitations section added, noting retrieval-stage metrics (Ragas, DeepEval) as future work

Supervisor response

The response was that hybridising techniques not improving QA accuracy is not a good finding, with direction to continue extending both draft and experimentation, introduce new research questions aimed at enforcing trustworthiness in medical QA, and set out goals and a timeline.

Assessment

The negative result is statistically sound but the interpretation is incomplete. One explanation not yet tested is that the backbone model is too weak for retrieval to help — `llama-3.3-70b-versatile` at 77.1% may simply be making errors that better context cannot fix. A stronger backbone would test whether the ceiling is the model or the method.

Plan for Week 7

1. Re-run the full evaluation with a frontier backbone model
2. Rebuild the knowledge graphs with the stronger model for higher-quality extraction
3. Investigate a larger, authoritative clinical knowledge graph rather than one extracted from benchmark documents
4. Extend evaluation to PubMedQA